DAY 8 | PRE-READ

A New Language That Never Waits

A light read before Day 8. About 15 minutes, no code required.

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Why read this?

So far you have built back ends in Python. Today you meet **JavaScript**, the language that runs inside every web browser, and with a tool called Node.js, on servers too. You will also learn the one big idea that makes JavaScript different: it never sits idle waiting for slow things to finish.

1. Meet JavaScript

If you know Python, JavaScript will feel familiar: it has variables, functions, lists (called **arrays**) and dictionaries (called **objects**). The main visible difference is that JavaScript uses curly braces { } instead of indentation.

Idea	Python	JavaScript
A list of things	list	array
Labelled information	dict	object
Print something	print()	console.log()
Group a block of code	indentation	{ curly braces }

2. Handing over instructions

When you ask a friend to "call me back when the parcel arrives", you give them an instruction to follow later. In JavaScript you often pass a function to another function, to be called later. That function is called a **callback**.

3. The coffee shop

Picture two coffee shops. In the first, the barista takes your order, makes the coffee, hands it over, and only then serves the next person. In the second, the barista takes your order, starts the machine, takes the next order, and calls your name when your coffee is ready. JavaScript works like the second shop. This is called **asynchronous** programming.

4. A receipt for something that is coming

When you order, you get a buzzer. It is a **promise** that your coffee will arrive. Eventually you get your coffee, or you are told something went wrong. Two friendly words, **async** and **await**, let you write "wait for the coffee, then add sugar" in a natural top-to-bottom style.

Buzzer status	Promise state
Waiting	pending
Buzzes: coffee ready	fulfilled

Buzzer status	Promise state
Staff apologises: out of milk	rejected

Key idea: start slow work, keep doing other things, and handle the result when it arrives.

5. Think about these before class

- Open any website, right-click and choose Inspect, then the Console tab. That is where JavaScript runs.
- Where else in daily life do you start something, do other things, and come back when it is ready?
- What should an app show you while it is waiting for data from the internet?

Before class, VS Code users: install Node.js LTS from nodejs.org.

Hajimemashite (はじめまして): nice to meet you, said when meeting someone new, like JavaScript